MIL-B-9946E
9 September 1987
SUPERSEDING
MIL-B-9946D
21 December 1977

MILITARY SPECIFICATION

BITS, SCREWDRIVER, GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers bits or bit assemblies used for installation and removal of recess drive fasteners (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-P-40 - Packaging and Packing of Hand Tools

(See supplement 1 for list of associated specification sheets)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Warner Robins Air Logistics Center, AFLC: ATTN: WR-ALC/MMEDTA, Robins AFB, GA 31096-5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5120
FSC 5130

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-1312	-	Fasteners, Test Methods
MS9006	-	Recesses - Cross, Low Torque Drive, Dimensions and Gage Dimensions For
MS33750	-	Recess, Hi-Torque and Conical (Connie) Hi-Torque Dimensions of Recess, Gage and Driver For
MS14191	-	Recess, Ribbed-Torq-Set Dimensions of Recess, Gage and Driver For
MS33781	-	Recess - Torq-Set, Dimensions of Recess, Gage, and Driver For

(Copies of specifications, standards, handbooks, drawings, publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

ASTM

ASTM A276	-	Standard Specification for Stainless and Heat-resisting Steel Bars and Shapes
ASTM A681	-	Standard Specification for Alloy Tool Materials
ANSI/ASTM E18	-	Standard Test Methods for Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1	-	Surface Texture (Surface Roughness, Waviness and Lay)
ANSI B107.4-1982	-	Driving and Spindle Ends for Portable Hand, Air and Electric Tools

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

AMS 6485	-	Steel Bars and Forgings 5.0Cr - 1.3Mo - 0.50V (0.38 - 0.43C) UNS T20811
AMS 6487	-	Steel Bars and Forgings 5.0Cr - 1.3Mo - 0.50V (0.38 - 0.43C) Premium Quality, Consumable Electrode Vacuum Melted

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(Application for copies should be addressed to the Society of Automotive Engineers, Inc., Warrendale, PA 15096).

NATIONAL AEROSPACE STANDARDS COMMITTEE

NAS 7101 - Bit, Screwdriver, Phillips, Specification For

(Application for copies should be addressed to the Aerospace Industries Association of America, Inc., 1725 DeSales Street, Washington, DC 20036).

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated specification sheets), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern (see 6.2).

3.2 Bid samples. Unless otherwise specified, each bidder shall furnish, with his bid, 25 samples of the bits he proposes to furnish under the contract for inspection as specified in 4.2.1. The procuring activity has the right to request additional samples from the production lot during the duration of the contract.

3.3 Material. Material shall be as specified on the applicable specification sheet. Recycled and reclaimed materials shall be used to the maximum extent possible.

3.3.1 Decarburization. All bits shall be free from decarburization as specified in ASTM A681.

3.4 Hardness. The hardness range of the bits shall be as specified on the applicable specification sheet.

3.5 Dimensions. Configuration of the driving blade shall be in accordance with the applicable specification sheet. The drive shank shall be as specified on the applicable specification sheet.

3.6 Surfaces. All surfaces shall be free of cracks regardless of depth. The driving blade and drive shank shall be free of discontinuities, inclusions, pitting or oxidation.

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3.6.1 Texture. The driving blade surfaces shall not be more than 63 microinches. All other surfaces shall not be more than 125 microinches, as specified in ANSI B46.1.

3.6.2 Protective finish and surface treatment. The bit shall be free of material permanently deposited on the surface (such as a plating or coating). A rust preventive shall be applied.

3.7 Performance.

3.7.1 Torque fatique. The bits shall be capable of withstanding values specified on the applicable specification sheet, in both the clockwise and counterclockwise directions, for 50 torque cycles.

3.7.2 Minimum ultimate torque. The bits shall be capable of withstanding, without permanent deformation or fracture, application of the minimum ultimate torque values specified in the applicable specification sheet.

3.8 Identification marking. Bits covered by this specification shall be marked for identification in accordance with the applicable specification sheet. Marking shall include the manufacturer's identification.

3.9 Workmanship. Bits shall be free of all imperfections which adversely affect the general appearance, function, or serviceability of the bits.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

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a. Bid sample inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 Bid sample inspection. When bid sample inspection is required (see 3.2), bid samples will be forwarded to the designated laboratory for testing (see 6.2). Failure of any sample to pass any of the tests specified shall be cause for rejection of the bid.

4.2.2 Quality conformance inspection. Quality conformance inspections shall consist of the tests specified in Table II. Sampling for inspection shall be in accordance with MIL-STD-105, except where otherwise indicated.

4.2.3 Certification. A certified test report showing that the manufacturer's product conforms to this specification shall be submitted with the bid sample.

4.3 Bid sample inspections. Bid sample inspections shall be as specified in Table I.

TABLE I. BID SAMPLE INSPECTIONS

INSPECTION	REQUIREMENT PARAGRAPH	TEST METHOD PARAGRAPH	NUMBER OF SAMPLE UNITS
Hardness	3.4	4.3.1	25
Material	3.3	4.3.2	2
Dimensions	3.5	4.3.3	23
Surfaces	3.6	4.3.4	23
Performance	3.7	4.3.5	23

4.3.1 Hardness. Each sample will be tested for hardness to macrohardness requirements in accordance with ANSI/ASTM E18 and as specified in the applicable specification sheets. The hardest and softest samples will then be sectioned longitudinally in such a manner as to bisect a blade and tested to determine compliance with the superficial hardness requirements as specified in the applicable specification sheet.

4.3.2 Material.

4.3.2.1 Decarburization. The two samples used to determine compliance with sectionalized hardness will be examined for decarburization. The cross section of each sample will be examined to determine compliance with decarburization requirements of 3.3.1.

4.3.2.2 Chemical requirements. The two samples used for decarburization inspection will be chemically analyzed to determine compliance with material chemical requirement, in applicable specification sheets.

4.3.3 Dimensions. Sample bits shall be inspected to determine compliance with the dimensional requirements specified in the applicable specification sheets.

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4.3.4 Surfaces. Sample bits shall be microscopically examined for surface defects and measured to determine compliance with smoothness requirement.

4.3.5 Performance.

4.3.5.1 Torque fatigue. Two sample bits that have passed dimensional and surface inspection and are representative of the lower shank hardness and minimum web thickness shall be subjected to torque fatigue testing using the appropriate test block specified in the applicable specification sheet. Samples not meeting dimensional requirements after the 50 cycle test shall be considered to have failed torque fatigue test.

4.3.5.2 Minimum ultimate torque. Two sample bits that have passed dimensional and surface inspections and are representative of the higher shank hardness and minimum web thickness shall be subjected to the ultimate torque test. The minimum ultimate torque load shall be as specified in the applicable bit specification sheet. Samples not meeting ultimate torque requirements as evidenced by permanent set or fracture shall be considered to have failed this minimum ultimate torque test.

4.4 Quality conformance inspection. Quality conformance inspection shall be as specified in Table II.

TABLE II. QUALITY CONFORMANCE INSPECTION

INSPECTION	REQUIREMENT PARAGRAPH	TEST PARAGRAPH	LEVEL	AQL
Hardness	3.4	4.5.3	S1	2.5%
Dimensions	3.5	4.5.1	II	1.0%
Surfaces	3.6	4.5.2	II	1.0%
Performance	3.7	4.5.4	S3	1.5%

4.4.1 Production lot. A production lot shall consist of bits which are the same type, class and size, fabricated from a single mill heat, heat treated and produced as one continuous run.

4.4.2 Inspection lot. An inspection lot shall consist of bits from a single production lot.

4.4.3 Sampling. Random samples for inspection and testing shall be selected from each inspection lot in accordance with MIL-STD-105. Inspection levels and AQLs shall be as specified in Table II.

4.5 Methods of inspection.

4.5.1 Dimensions. Sample bits shall be inspected for dimensional accuracy using conventional measuring equipment (a micrometer or calipers) and a gage as specified (see 4.5.1.1). Noncompliance with the dimensions specified on the applicable specification sheet or drawing shall constitute failure.

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4.5.1.1 Gage. Sample bits shall meet the "Go - No Go" criteria established by gages for that particular driver configuration as specified in the applicable specification sheet or when the applicable gage is not of the "Go - No Go" type, the sample bits shall be inspected after the gage has been fitted to a master gage and a determination made that the inspection gage has not worn excessively. Noncompliance with the gage shall constitute rejection of the lot.

4.5.2. Surfaces. Sample bits shall be microscopically examined at a magnification of not less than 10X. Failure to meet the requirement shall be cause for rejection (see 3.6).

4.5.2.1 Smoothness. Sample bits shall be inspected in accordance with ANSI B46.1 for surface smoothness. Any readings exceeding the required limits shall be cause for rejection.

4.5.2.2 Surface treatment. Sample bits shall be visually inspected for conformance to the requirements of 3.6.2.

4.5.3 Hardness. Sample bits shall be tested in accordance with ANSI/ASTM E18.

4.5.4 Performance.

4.5.4.1 Torque fatigue. Sample bits shall be tested for torque fatigue using the appropriate test blocks as specified in the applicable specification sheet. There shall be no debris present in the recess during the test. Torque fatigue load shall be applied alternately for 50 cycles. Testing shall be in accordance with MIL-STD-1312, Test 25 except as follows:

a. Torque load: in accordance with applicable specification sheet.

b. Torque direction: alternately clockwise and counterclockwise.

c. End load: 20 to 20.5 pounds.

d. A cycle torque loading consists of once in the clockwise plus once in the counterclockwise direction.

4.5.4.2 Minimum ultimate torque. Sample bits shall be tested for the minimum ultimate torque using test blocks as specified in the applicable bit specification sheet. The torque load shall be applied with a calibrated torquing device of ±5 percent accuracy. Dead weight load the torque testing machine using an axial end load of 20 to 20.5 pounds to maintain full engagement with test block recess during the test. The driving blade axis shall be aligned with the recess axis within 3°. The minimum ultimate torque shall be applied once in the clockwise direction. Retest using a new bit. The tested bit shall not be used for further tests. A new bit shall be used with the minimum ultimate torque applied once in the counterclockwise direction. Retest using a new bit. The tested bit shall not be used for further tests.

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4.6 Inspection of packaging. The sampling and inspection of the preservation, packing and container marking shall be in accordance with packaging requirements (see 5.1).

5. PACKAGING

5.1 Packaging requirements. Unless otherwise specified, packaging shall be in accordance with PPP-P-40, Level "C" (see 6.2).

5.2 Marking. Unless otherwise specified, marking shall be in accordance with MIL-STD-129 (see 6.2).

6. NOTES

6.1 Intended use. Bits shall be capable of being used with selected hand and power wrenches in the installation and removal of recess drive fasteners.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification and applicable specification sheets (see 3.1).

b. Applicable specification sheet part number.

c. Additional tests required.

d. Preservation packaging and packing (see 5.1).

e. Bid sample inspection is required, designated testing laboratory (see 4.2.1).

f. For marking requirement (see 5.2).

6.3 Definitions.

a. Bits. A bit is a tool with an appropriately shaped driving blade used to apply torque to the recess drive element of a screw or bolt.

b. Driving blade. The driving blade is that portion of the bit that engages the recess drive element in the screw or bolt.

c. Drive shank. The drive shank is the portion of the bit that engages the wrench.

d. Laps. Laps are material discontinuities (folded metal) that do not cross grain boundaries.

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e. Cracks. A crack is a clean, crystalline break passing through grain or grain boundary.

f. Plating and coating. Plating is electro-chemical deposition of material (such as cadmium, chromium, zinc, etc.); coating is a mechanical deposition of material (such as paint, organics, suspension containing metals, etc.). Coating and plating do not include chemical processes such as anodize, black oxide or black phosphate (see 3.6.2).

6.4 Key word listing.

Bit, Screwdriver
Bit, Screwdriver, External Hex Shank
Bit, Screwdriver, Hi-Torque
Bit, Screwdriver, Manually Driven
Bit, Screwdriver, Phillips
Bit, Screwdriver, Phillips, Manually Driven
Bit, Screwdriver, Phillips, Power Driven
Bit, Screwdriver, Phillips, Ribbed
Bit, Screwdriver, Power Driven
Bit, Screwdriver, Ribbed
Bit, Screwdriver, Six Spline
Bit, Screwdriver, Slot
Bit, Screwdriver, Standard and Long,External Hex Shank
Bit, Screwdriver, Torq-Set
Bit, Screwdriver, Torq-Set, Manually Driven
Bit, Screwdriver, Torq-Set, Power Driven
Bit, Screwdriver, Torq-Set, Ribbed

6.5 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - GL
Navy - AS
Air Force - 99

Review Activities:
Army - AV
Air Force - 80
DLA - IS

User Activities:

Civil Agencies Coordinating Activities:
GSA - FSS

Preparing Activity:
Air Force - 84

Agent:
DLA - IS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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